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BARCELONA ZETTASCALE LAB

Bringing GNU/Linux to work with experimental RISC-V cores

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Summer School on RISC-V Hardware-Software Co-Design Training
Namal University, Mianwali

Outline

- The BZL project
- Hardware emulation platform
- Porting/adapting software
- Benchmarking
- Conclusions

The BZL project



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Barcelona Zettascale Lab project

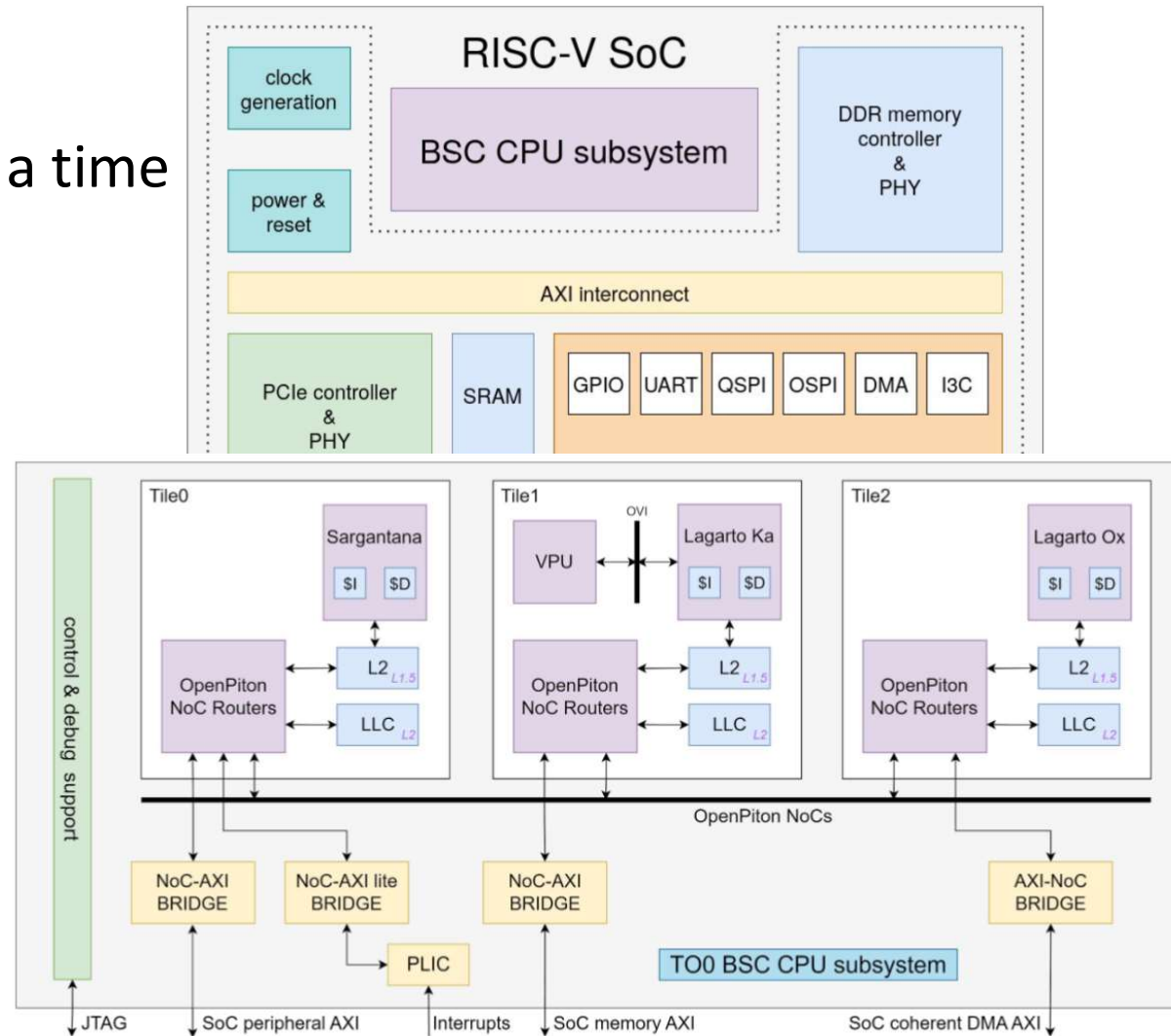
- Period: Dec '22 to June '26 (3.5 years)
- Build first BSC-only HPC multicore design based on RISC-V
 - Out-of-order core, including vector processing
 - HPC cache hierarchy
- Two test chips with Intel Foundry Systems
 - Intel provides uncore devices (memory controller, PCIe,...)
- Test-Chip #1: end of 2024
- Test-Chip #2: 2026

BZL Team

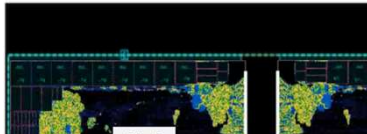
- PI: Mateo Valero
- HW coordination: Miquel Moretó
 - Arch modeling: Adrià Armejach
 - Microarch: Luis Plana
 - Verification: Oscar Palomar
 - FPGA prototyping: Behzad Salami
 - Physical design: Francesc Moll
 - PCB and bring-up: Francesc Moll
- SW coordination: Xavier Teruel
 - SDV: Filippo Mantovani
 - OS: Xavier Martorell
 - Compiler: Roger Ferrer
 - Analysis tools: German Llorca
 - Libraries, frameworks: Rosa Badia
 - Applications: Xavier Teruel

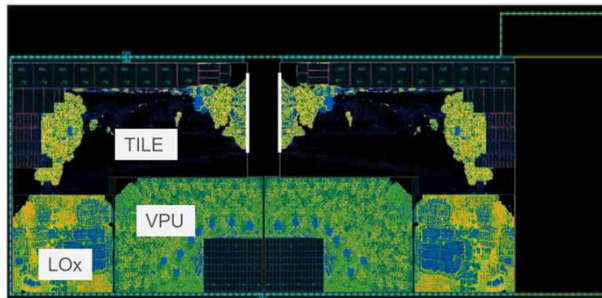
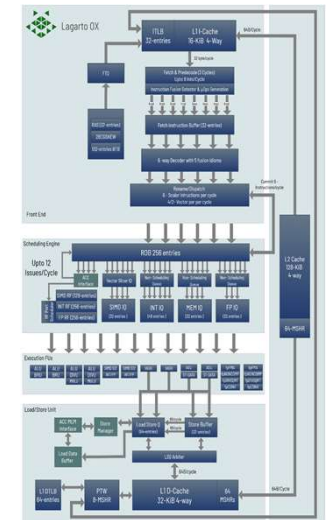
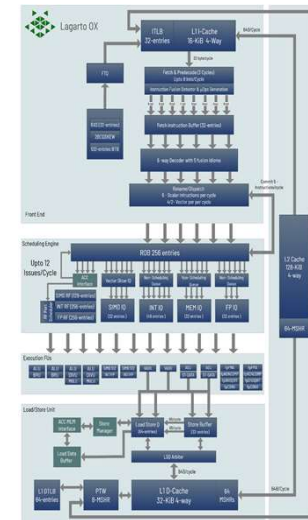
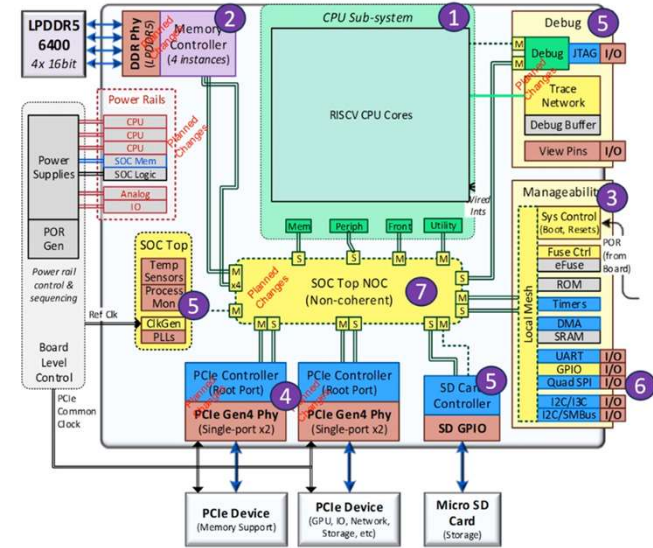
Hardware Test-Chip #1

- 3 cores – only one enabled at a time
 - Sargantana
 - In-order, single issue
 - rv64imafd
 - Lagarto Ka + VPU
 - Out-of-order, dual-issue
 - rv64imafdv v 1.0 VPU
 - Lagarto Ox
 - Out-of-order, 6-way issue
 - Rv64imafdc
- Tech. node: Intel3



Hardware Test-Chip #2

- 2 cores – enabled as an SMP platform
 - **Lagarto Ox + VPU**
 - Out-of-order, 6-way issue
 - rv64imafdcv
 - Tech. node: Intel 18A
 - 12x4 mm², ~5mm² (BSC)
 - Similar uncore environment (to TC#1)
 - Goal for the future:
complete the full **RVA23** profile
- 



Test Boards

- Board for Test-Chip #1
 - Single PCIe slot



- Board for Test-Chip #2
 - 4 PCIe slots
 - USB ports

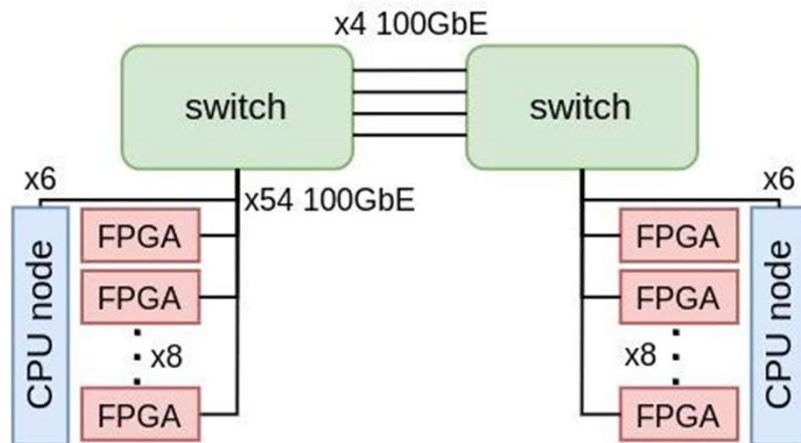


Hardware emulation platform



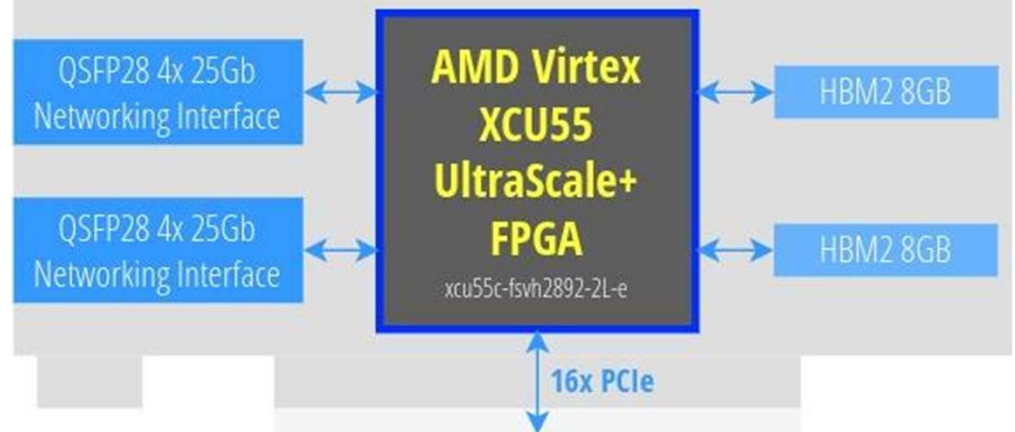
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MEEP cluster architecture



- 96 Alveo U55C FPGAs
- 12 Intel Xeon Gold 6330 CPUs with 256GB RAM
- 2 Nvidia SN4600C 100GbE switches

AMD Alveo U55C high performance compute card

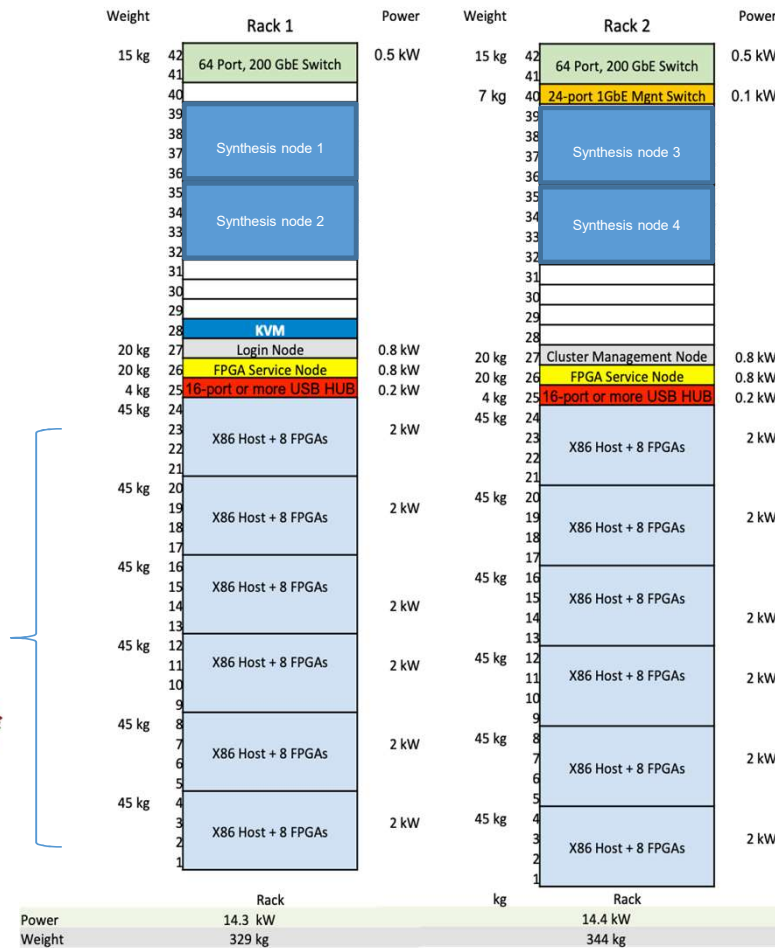


MEEP cluster architecture

- 1 “login” node
- 4 “synthesis” nodes
- 12 “compute” nodes
 - 8 u55c PCIe FPGA boards per node
- 2 switches
 - 96 FPGAs
 - 12 nodes

Private, “MPI” network

x48 Alveo u55C



x48 Alveo u55C



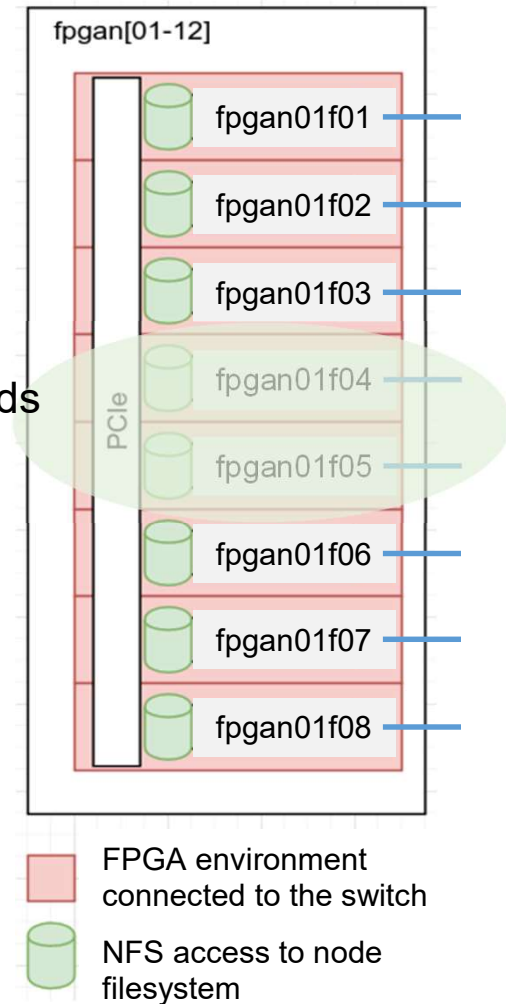
MEEP cluster architecture

- Multiuser environment
 - 1 node – 1 user (initially)
 - 1 FPGA – 1 user (preferred)
- BSC Operations department implemented the support
 - Based on the SLURM application scheduler
 - Allocating full nodes to a single user (main partition)
 - Allocating any number of FPGAs per node to each user (single partition)
 - Max 96 users with a single FPGA
 - Based on a SLURM plugin, adapted from GPU support

Using SLURM to allocate FPGAs

- On the “login” node, SLURM “single” partition
 - `--partition=single`
 - `--gres=gpu:u55c:2`
 - gpu means fpga (plugin adapted)
 - u55c => type of the device
 - 2 => how many requested
 - `ASSIGNED_FPGA = 4,5`
 - Environment variable with the FPGA IDs

2 FPGA boards
allocated



Reconfiguring the FPGAs

- On the SLURM-allocated node, e.g., fpgan01

- module load fpga-tools-v2

[fpgan01 ~]\$ ft2-loader -help

ft2-loader [args ...] - The new tools/utls to work with fpgas.

- b, **--bitstream** <file> Loads a bitstream using Vivado
- o, **--osbi** Loads the firmware image
- f, **--fs** Loads the file system image
- k, **--kernel-fs** Loads the kernel-fs image
- auto** <dir> Gets -b, -o, -f and -k from the directory <dir>
- c, **--config** <config> Shortcuts for load addrs.

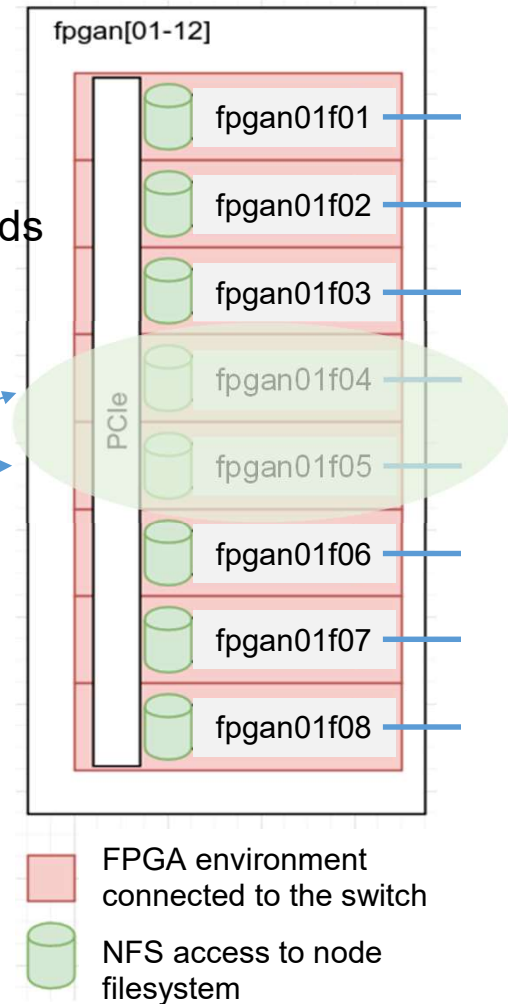
Available configs => cr, sa-sa, sa-ox, nobootroom, tc2, **komodo**, acme, **spiton**, dare

--parallel Enables parallel execution of some tasks

-l, **--fpga-list** <list> FPGA list to work with (makinote only) [1-8]

<list> is a separated-comma list, eg=4,5

2 FPGA boards allocated



Porting/adapting software



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Porting/adapting software

- Firmware
 - OpenSBI, U-boot, DeviceTree
 - GRUB, Tianocore – UEFI support
- Operating System
 - GNU Linux kernel 6.8 => 6.12.11 => 6.16.12
- Root filesystem
 - Based on Buildroot (buildroot.org)

- BootROM (on bitstream) invokes OpenSBI

 -- Komodo Platfor --

Config: 1tlox_2lvpu_20mhz

#X-Tiles: 1

#Y-Tiles: 1

Cores: 1

Lox Config: 2ways

Core Freq: 20 MHz

L1I Size / Assoc: 16 kiB / 4

L1D Size / Assoc: 16 kiB / 4

L15 Size / Assoc: 64 kiB / 8

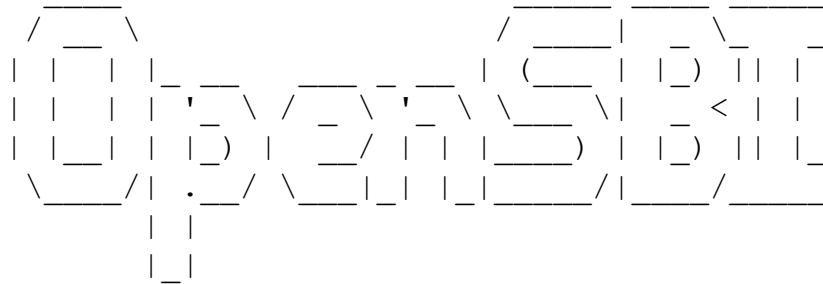
L2 Size / Assoc: 256 kiB / 8

L15/L1D Cacheline size 64



OpenSBI

OpenSBI v1.7



Platform Name : Generic

Platform Features : medeleg

Platform HART Count : 1

Platform IPI Device : aclint-mswi

Platform Timer Device : aclint-mtimer @ 156250Hz

Platform Console Device : uart8250

...

Firmware Base : 0x8000000000

Firmware Size : 317 KB

Firmware RW Offset : 0x40000

Firmware RW Size : 61 KB

Firmware Heap Offset : 0x46000

Firmware Heap Size : 37 KB (total), 2 KB (reserved), 12 KB (used), 23 KB (free)

Firmware Scratch Size : 4096 B (total), 408 B (used), 3688 B (free)

Runtime SBI Version : 3.0

Standard SBI Extensions : time,rfnc,ipi,base,hsm,pmu,dbcn,fwft,legacy,sse

Experimental SBI Extensions : none

OpenSBI

- Brings RISC-V harts online
- Prepares the switch to Supervisor mode
- Switches to Supervisor by jumping to the payload
 - U-Boot
 - UEFI Boot
 - Linux kernel

```
Domain0 Name      : root
Domain0 Boot HART  : 0
Domain0 HARTs      : 0*
Domain0 Region00   : 0x0000000100802000-0x0000000100802fff M: (I,R,W) S/U: (R,W)
Domain0 Region01   : 0x00000008000040000-0x0000000800004ffff M: (R,W) S/U: ()
Domain0 Region02   : 0x00000008000000000-0x0000000800003ffff M: (R,X) S/U: ()
Domain0 Region03   : 0x0000000e000500000-0x0000000e00057ffff M: (I,R,W) S/U: ()
Domain0 Region04   : 0x0000000e200000000-0x0000000e20ffffff M: (I,R,W) S/U: (R,W)
Domain0 Region05   : 0x00000000000000000-0xffffffff M: () S/U: (R,W,X)
Domain0 Next Address : 0x00000008000200000
Domain0 Next Arg1    : 0x0000000800a200000
Domain0 Next Mode     : S-mode
Domain0 SysReset      : yes
Domain0 SysSuspend    : yes

Boot HART ID        : 0
Boot HART Domain     : root
Boot HART Priv Version : v1.12
Boot HART Base ISA    : rv64imafdcbv
Boot HART ISA Extensions : sscofpmf,sstc,zicntr,zihpm,smcntrpmpf,sdtrig
Boot HART PMP Count   : 16
Boot HART PMP Granularity : 6 bits
Boot HART PMP Address Bits : 38
Boot HART MHPM Info    : 29 (0xffffffff8)
Boot HART Debug Triggers : 0 triggers
Boot HART MIDELEG      : 0x0000000000000222
Boot HART MEDELEG      : 0x0000000000000b109
```

Linux kernel

- Completes de Supervisor environment
 - E.g. DeviceTree description of the memory

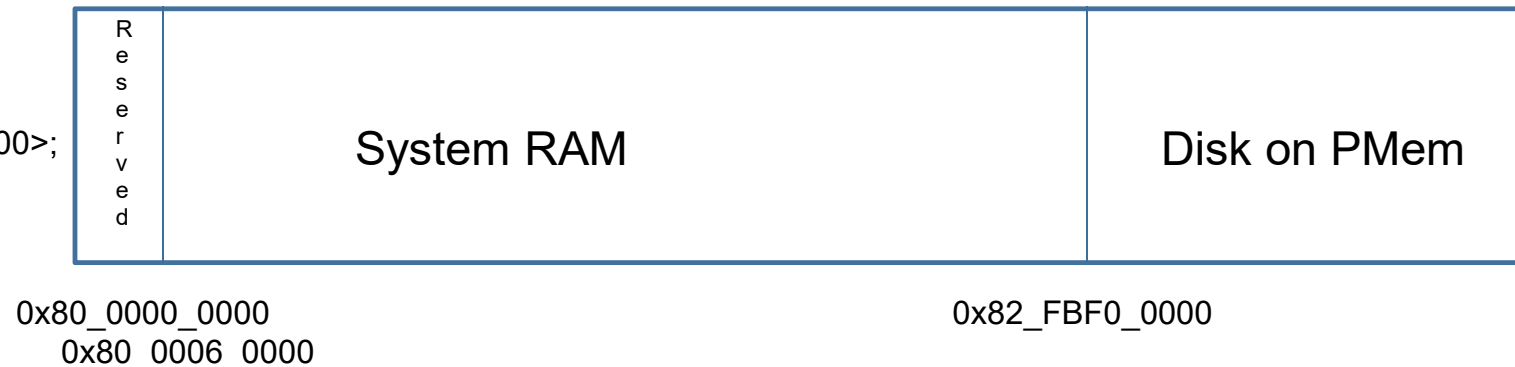
```
memory@8000000000 {
    u-boot,dm-pre-reloc;
    device_type = "memory";
    reg = <0x80 0x00 0x03 0x3bf00000>;
};
```

```
pmem@833bf00000 {
    compatible = "pmem-region";
    volatile = [00];
    reg = <0x83 0x3bf00000 0x00 0xc0100000>;
};
```

```
reserved-memory {
    #address-cells = <0x02>;
    #size-cells = <0x02>;
    ranges;

    mmode_resv1@80,0 {
        reg = <0x80 0x00 0x00 0x40000>;
        no-map;
    };

    mmode_resv0@80,40000 {
        reg = <0x80 0x40000 0x00 0x20000>;
        no-map;
    };
};
```



Physical memory map

Porting/adapting software

- Buildroot
 - Configured with a Linux-style “menuconfig”
 - Organizes the compilation of all components
 - The **compiler** itself – host native and cross-compiler versions - GCC
 - We have incorporated the generation of the compiler for the target
 - **U-Boot**
 - The GNU **Linux** kernel
 - The **Init** process – busybox, System V Init, or SystemD
 - The utilities for the **root filesystem**
 - Libraries – glibc to X11, shells, utilities, servers, small applications...
 - Compatible enough to behave close to Ubuntu or other Linux distributions

Porting/adapting software

- Buildroot also supports generating a Linux kernel adapted to the hardware, and able to boot Linux distributions. We have tested:
 - Ubuntu, Debian
 - Fedora
 - Arch
 - Rocky Linux
- One can get a raw filesystem image (Ubuntu...) and try

Porting/adapting software

- Login credentials use LDAP from host servers
 - Same user/password
 - Possibility to install SSH keys for access without password
- RISC-V node mounts HOME directory by NFS
 - Users can cross-compile applications on Intel host...
 - ... and immediately run them on RISC-V

Porting/adapting software

- We have contributed on the detection of several hardware issues
 - PLIC – Peripheral Interrupt Controller behaviour
 - Vector instructions generating wrong results
 - Incorrect behaviour of a combination of Compressed instructions
 - Generating the proper DeviceTree for the PMU – hardware counters
 - Missing features or attributes in the DeviceTree
 - Currently being able to use the Ethernet driver from AMD, instead of the internal version we were using from FORTH

Benchmarking



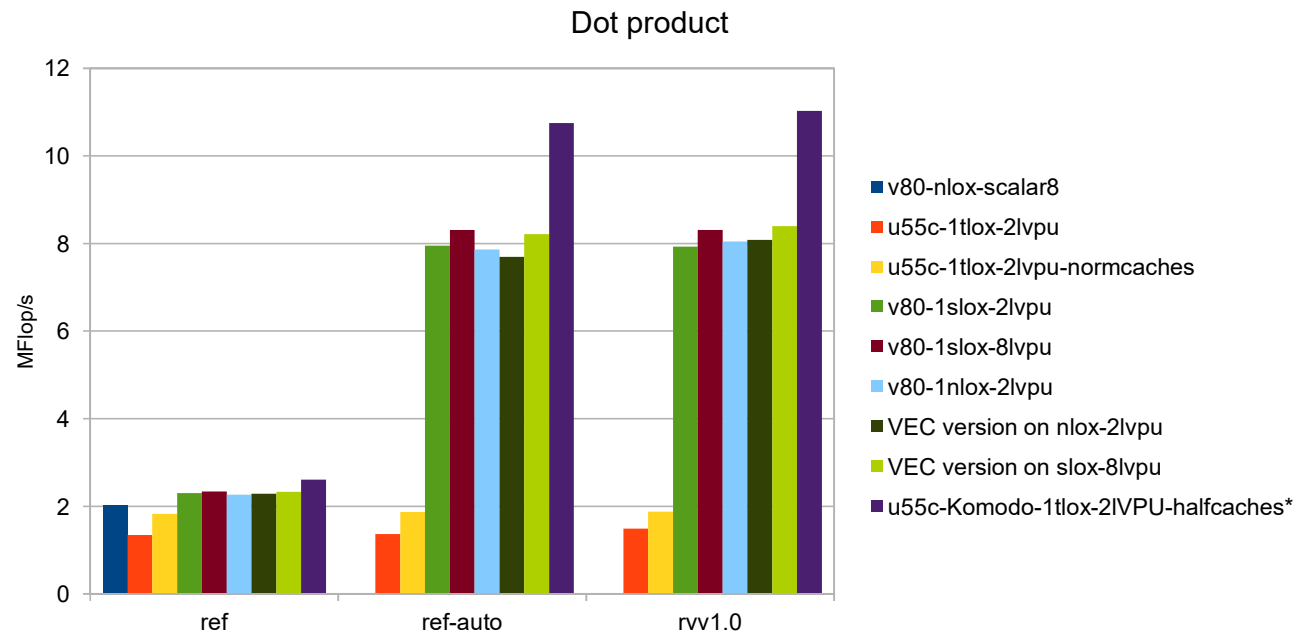
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Benchmarks tested

- Software Development Vehicle – Cross-Architecture Benchmarks
 - Dot product AXPY
 - SpMV GEMV
 - GEMM FMA
 - Conj-Grad Jacobi
 - STREAM-lite
- HPL Linpack, HPCG, HPCC
- NEMO dwarfs
- and many more

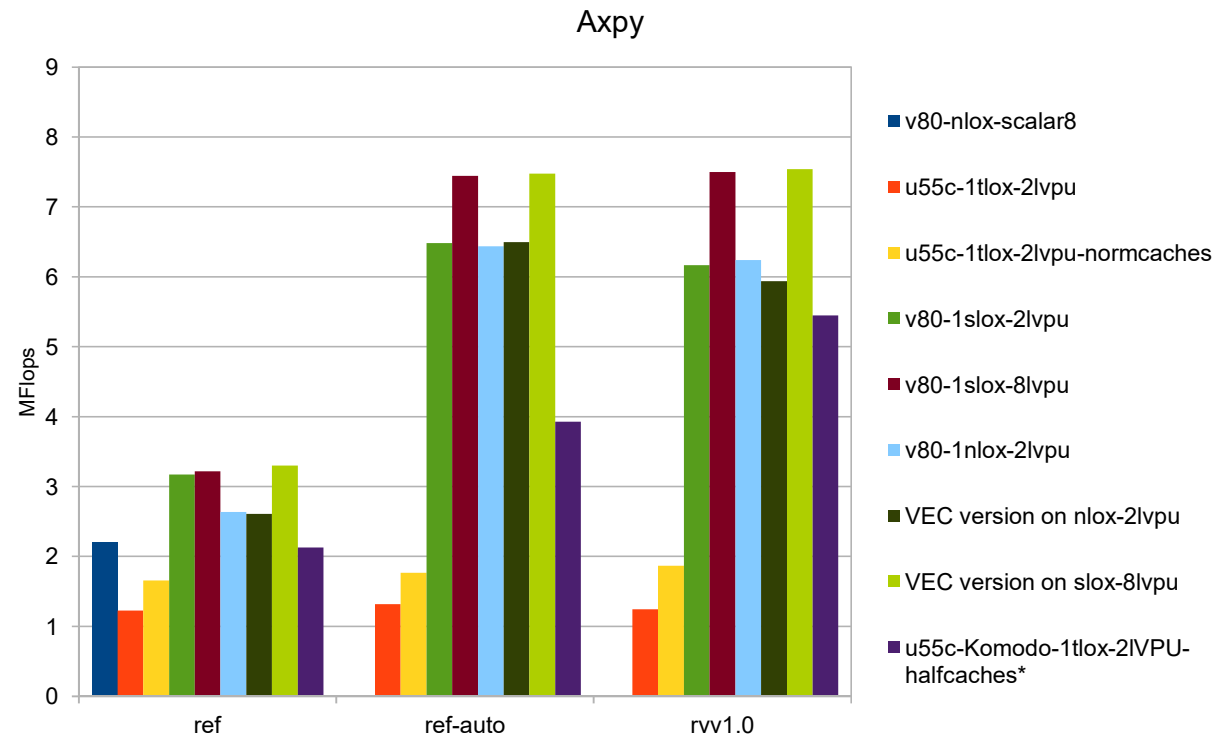
Benchmarking

- Dot product
 - Comparing FPGAs v80 – u55c
 - Different Ox cores
 - Normal
 - Small
 - Tiny
 - Different VPU units
 - Not available
 - 2 lanes
 - 8 lanes



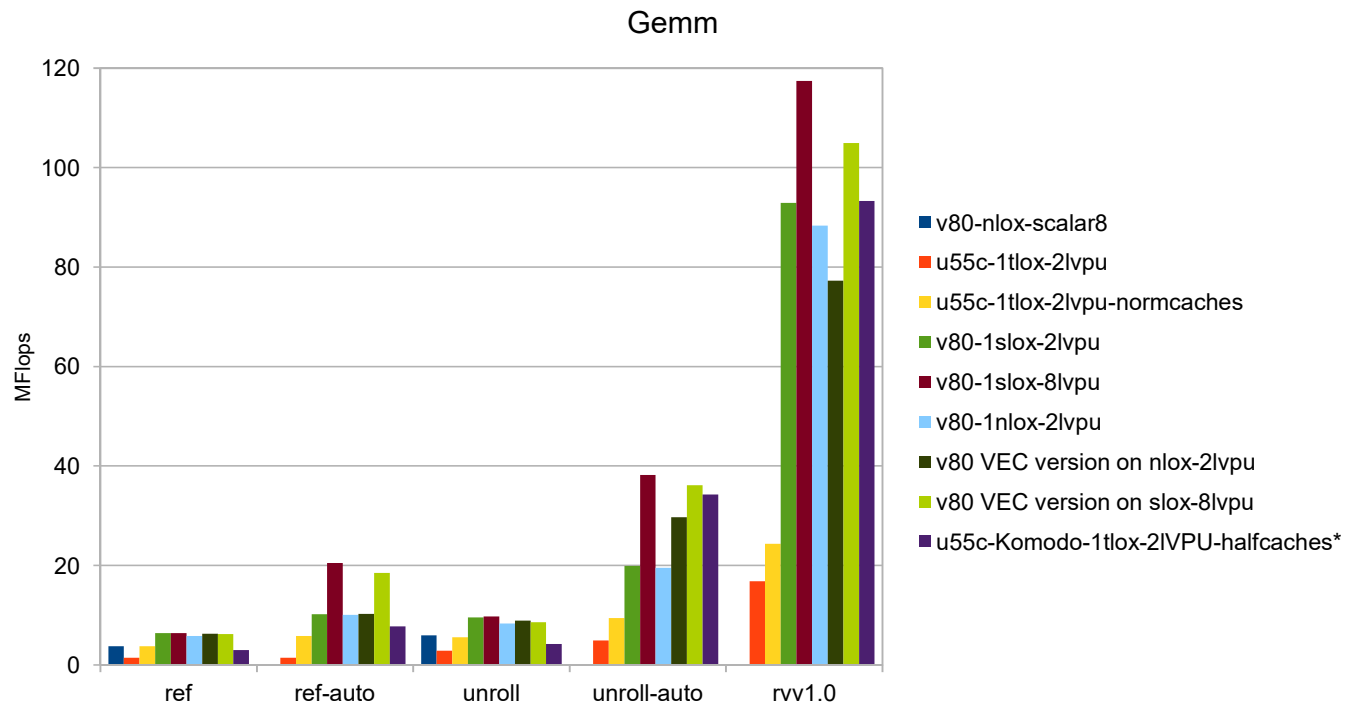
Benchmarking

- AXPY
 - Memory bound, limits vector performance



Benchmarking

- GEMM – matrix multiply
 - Compute bound, this is the one showing good vector performance



Conclusions



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Conclusions

- Adapting the firmware and OS for a moving hardware platform is a challenge
- On the RISC-V architecture, the “device-tree” is the perfect interface between the hardware and the software teams
- Current software open-source tools can be adapted to the RISC-V architecture in a reasonable amount of time
- Completed an environment for booting Linux distributions and support benchmarking of the hardware



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Bringing GNU/Linux to work with experimental RISC-V cores

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<https://bzl.es/en/>



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